Zachary Newsom

(509) 392-9949; zachary.newsom1@gmail.com; LinkedIn; GitHub

EXPERIENCE / PROJECTS

Power System Planning Intern – Avista Utilities *May 2024 – Aug 2024*

- Wrote **system upgrade** studies/recommendations (for both transmission and distribution level planning), **generator/load interconnection** studies and created capacity-need estimates for different development types.
- Learned about the economic and technical complexities around planning for future electricity demand in Eastern WA/North ID. **PowerWorld** and **Synergi** for powerflow analysis, performed load growth sims and did studies on mitigating and solving future issues on the grid. Interfaced with **substation, protection and operations** teams.

Capstone – Design, Model, and Test Advanced Communication and Sensing Electronics at Cryogenic Temperatures – Boeing (sponsor), WSU Systems-on-Chip Lab (faculty) *Sep 2024 – May 2025*

- Objective was to design, model, and test a 40–44 GHz **FMCW radar** system optimized for cryogenic environments via the ICE-T test-bed. Acted as the sole **PCB drafter/designer**, using **Altium** to create boards for **cryo-characterizing** individual components and the RADAR board itself. Learned about **RF PCB design** guidelines.
- Developed **MATLAB/Simulink** RADAR simulations, contributed to the design of a **high-gain PCB patch antenna** using **ADS simulation** and Altium, and integrated a commercial **RFSoC** (ZCU111) for signal gen and processing.
- Characterized and tested commercial electronic components at **cryogenic temperatures** (down to 4K) to evaluate performance gains from **thermal noise reduction** and find their functional breakdown temperature.

Test Engineer Intern – Protective Relay Manufacturing - Schweitzer Engineering Laboratory *Sep 2023 – Jun 2025*

- Worked in the 651R Recloser / 2431 Regulator controls manufacturing department, learned about the process engineering going into a high-throughput, **complex manufacturing floor**. Designed and maintained test cabinets and custom scripts for technicians to functionally test products before shipping. Helped with **support requests**.
- Learned how to **troubleshoot** products at a **cabinet and board level** when manufacturing faults were detected.
- Programmed **custom instrument control code**, for our in-house manufacturing test scripting language, that implemented externally-designed cable/harness testers (CAMI Research) as scriptable instruments. Learned about COM and build systems, created extensive documentation, implemented use on the manufacturing floor.

Indie Game Developer / Game Development Club Founder and Teacher *Feb 2023 – May 2025*

- **Founder** and **former president** of WSU's Game Development Club, which has grown to be one of the largest new clubs on campus, with 100+ participating in our yearly game jams. Experience in **advertising and growing a club**, as well as hosting and funding **large events**. Taught workshops on pixel art, 3D art, coding, and game design.
- **Self-taught indie game developer** with experience programming in a variety of engines/frameworks and languages (primarily with Unity, Godot and **C++/OpenGL**). Experienced in digital art creation with focus towards video game assets. Programmed numerous small-scale, indie video games since I was a teen (**8+ years coding**).

EE/CS Lead - Crimson Robotics (WSU BattleBots/Combat Robotics Team) *Sep 2021 – Jul 2023*

- Experience in complex robot electronics as the former EE/CS lead of WSU's "BattleBots" team; made custom PCBs for a **LiPo battery monitoring system**, a **ESP32-based** motherboard, and FrSky-based telemetry.
- Configured FOC-based **ESCs**; power-rated components, motors, wiring; used **SolidWorks** for modeling robot.

EDUCATION

Washington State University (Pullman, WA) **August 2021 – May 2025**
Bachelor of Science, Electrical Engineering - Systems Track ***GPA:*** *3.62*

Extracurricular: Game Development Club, Crimson Robotics, IEEE, HAM Radio Club, Climbing Club, WSU ESIC

SUMMARY OF QUALIFICATIONS

- Electrical Engineering major with strong practical skills in the field. **HAM/Amateur radio General License!**
- Electronics troubleshooting/repair; operated oscilloscopes, multimeters, power supplies, VNAs, logic analyzers
- ***Languages***: C/C++, MATLAB, ARM Assembly, Verilog, VHDL, C#, Rust, Python, Go, JavaScript (React)
- ***Relevant Software*****:** Altium Designer, KiCad, LTSpice, Cadence Virtuoso, AMD Vivado/Vitis, Keysight ADS, PowerWorld, Synergi Electric, SolidWorks, Fusion 360, MATLAB/Simulink, LabVIEW, Visual Studio
- ***Coursework:*** Power Electronics, Intro to Control Systems, Digital Communication Systems, Digital Signal Processing, RF and Microwave Circuits/Systems, Computer Networks/Security, Digital Logic, Microprocessors
  - ***Class Projects*****:** Simulation of a Three-Phase SPWM Inverter Driving an Induction Motor with V/f Control, AM Transmitter & Receiver, Over-Radio AX.25-Based BBS System, Verilog-Based VGA Display Controller